VIRAMGAM · GUJARAT · INDIA

# Establishing Modern Standards in Tube Manufacturing

A precision-focused tube manufacturer delivering premium cold-finished seamless stainless steel tubes and pipes for industrial, instrumentation, heat-exchanger, and ultra-high-purity applications — backed by rigorous process control and international-grade documentation.

| 01 · CUSTOM DEVELOPMENT | 02 · QUALITY SYSTEMS | 03 · PROCESS PRECISION | 04 · GLOBAL SUPPORT |
|---|---|---|---|
| Bespoke OD, wall thickness, and alloy chemistry engineered for demanding duty. | ISO 9001, 14001 & 45001 certified management with EN 10204 3.1/3.2 MTCs. | Straight, corrugated, U-bend, coil, electropolished, and cleanroom UHP lines. | Export-ready documentation and logistics across 10 critical global industries. |

DIMENSIONAL RANGE
**OD 3.00 – 60.30 mm**
WT 0.60–8.00 mm · Up to 12 m & 250 m Coils

GLOBAL STANDARDS
**ASTM · ASME · EN · DIN**
ASME BPE · SEMI F19/F20 · EN 10204 3.1/3.2

CERTIFIED MANAGEMENT
**ISO 9001 · 14001 · 45001**
29 Years of Founder Industry Experience

PAGE 02 · CORPORATE PROFILE & STRATEGIC FOUNDATION

# 29 Years of Leadership in Precision Seamless Tube Manufacturing

AV-Tech Tube Industry is an ISO-certified manufacturer of cold-finished seamless stainless steel tubes and pipes located in the Ahmedabad industrial corridor of Viramgam, Gujarat, India. By integrating advanced cold-drawing and bright-annealing metallurgy with strict quality protocols, AV-Tech Tube Industry serves domestic and international markets across process piping, instrumentation, heat-transfer, and ultra-high-purity applications.

## Mission, Vision & Core Operational Pillars (avtechtube.com/about)

| Strategic Pillar | Corporate Commitment & Execution Standard |
|---|---|
| **Our Mission** | To establish a new benchmark in global tube manufacturing by delivering products that exceed international quality standards while offering the agility to support bespoke technical requirements. |
| **Our Vision** | To be the premier international partner for engineered tube solutions, globally recognized for uncompromising quality, operational reliability, and deep metallurgical expertise. |
| **01 · Custom Development** | Tailored outer diameters (3.00–60.30 mm), wall thicknesses (0.60–8.00 mm), bend geometries, and controlled-chemistry alloys engineered to exact project specifications. |
| **02 · Robust Quality Systems** | Rigorous control protocols from raw material mother-hollow verification through cold drawing, bright annealing, non-destructive testing, and EN 10204 3.1/3.2 certification (Quality Systems). |
| **03 · Strategic Optimization** | Continuous process refinement across cold-pilgering, drawing, pickling, passivation, and electropolishing to achieve superior surface finishes and dimensional precision. |
| **04 · Comprehensive Support** | Dedicated techno-commercial responsiveness covering specification review, feasibility evaluation, export-grade seaworthy or cleanroom packaging, and transparent lead-time tracking. |

## Founder's Techno-Commercial Advantage: Mr. Minesh Vyas

With **29 years of dedicated experience** in the stainless steel tube industry, Founder Mr. Minesh Vyas brings a rare combination of shop-floor metallurgical mastery and global commercial insight to AV-Tech Tube Industry — overseeing raw material procurement, tool design, machine optimization, and global market expansion.

> *"We aren't just supplying stainless steel tubes; we are engineering long-term reliability for our partners."* — MR. MINESH VYAS, FOUNDER

## Four Reasons Global Buyers Partner with AV-Tech Tube Industry

| Partnership Advantage | How It Protects Client Operations |
|---|---|
| **01 · Technical Authority** | Deep engineering expertise in cold-drawn seamless tube metallurgy, wall-thickness tolerance control, corrosion resistance, and application-specific performance. |
| **02 · Client-Centric Solutions** | Direct collaboration with EPC contractors, OEMs, and plant engineers to deliver purpose-built tubing lengths, U-bend radii, and internal bore finishes rather than generic stock. |
| **03 · Operational Excellence** | Strict process discipline, workplace safety (ISO 45001), and environmental stewardship (ISO 14001) that guarantee repeatable quality and predictable on-time outcomes. |
| **04 · Global Readiness** | Export-aligned documentation workflows, multi-language communication streams, and full compliance with ASTM, ASME, EN, DIN, ASME BPE, and SEMI standards. |

**Manufacturing Hub:** Viramgam Industrial Corridor, Ahmedabad, Gujarat, India · **Certifications:** ISO 9001:2015 · ISO 14001:2015 · ISO 45001:2018

# Straight, Corrugated & “U” Bend Seamless Tubes for Process & Thermal Duty

AV-Tech Tube Industry manufactures cold-finished Stainless Steel Seamless Tubes and Pipes with zero longitudinal weld seam. Cold-pilgered and cold-drawn from certified mother hollows, our tubes deliver uniform grain structure, isotropic mechanical strength, and high pressure integrity across extreme temperature and corrosive cycles.

## Straight, Corrugated & “U” Tube Specifications

| Product Line | Dimensional Range & Codes | Metallurgical Execution & Industrial Service |
|---|---|---|
| Seamless Straight Tubes & Pipes | **OD:** 6.35 – 50.80 mm (up to 60.30 mm)<br>**WT:** 0.60 – 8.00 mm<br>**Pipe:** 1/4" – 4" NB (SCH 5S–160S)<br>**Length:** Up to 12 meters | Round hollow sections supplied in straight lengths for high-pressure process piping, refinery utilities, hydraulic lines, and boiler service per ASTM A312/SA312, A213/SA213, A269, EN 10216-5, and DIN 17456/17458. |
| Seamless Corrugated Tubes | **OD:** 6.35 – 50.80 mm<br>**Nominal Size:** DN12 – DN50<br>**Standards:** ASTM A213 / A269 / EN 10216-5 | Rigid seamless stainless steel tubes formed with precision helical corrugations that break boundary-layer laminar flow, inducing turbulence that enhances convective heat transfer by **2 to 3 times** while resisting surface fouling. |
| Seamless “U” Tubes | **OD:** 9.52 – 50.80 mm<br>**WT:** 0.70 – 8.00 mm<br>**Standard:** ASTM / ASME SA213 | Precision cold-bent U-tubes for shell-and-tube heat exchangers, power plant condensers, and reboilers. Supplied with controlled wall thinning at the extrados, bend-zone stress-relief annealing, and hydro-testing. |

## Reference Weight Schedule: Seamless “U” & Straight Tubes (kg/m)

Standard theoretical unit weights (kg/m) for austenitic stainless steel seamless tubes across standard heat-exchanger and process outer diameters (U-Tube Technical Schedule):

| Outer Diameter (mm) | 0.70 mm WT | 1.00 mm WT | 1.65 mm WT | 2.00 mm WT | 2.60 mm WT | 3.00 mm WT |
|---|---|---|---|---|---|---|
| **9.52 mm (3/8")** | 0.154 kg/m | 0.213 kg/m | 0.330 kg/m | 0.390 kg/m | 0.488 kg/m | 0.546 kg/m |
| **12.70 mm (1/2")** | 0.210 kg/m | 0.293 kg/m | 0.444 kg/m | 0.540 kg/m | 0.685 kg/m | 0.773 kg/m |
| **19.00 mm (3/4")** | 0.320 kg/m | 0.450 kg/m | 0.696 kg/m | 0.850 kg/m | 1.066 kg/m | 1.200 kg/m |
| **25.40 mm (1.0")** | 0.432 kg/m | 0.610 kg/m | 0.952 kg/m | 1.170 kg/m | 1.482 kg/m | 1.680 kg/m |
| **50.80 mm (2.0")** | — | — | 1.968 kg/m | 2.440 kg/m | 3.133 kg/m | 3.585 kg/m |

## Dimensional Tolerance & End-Preparation Capabilities

All straight, corrugated, and U-bend tubes are supplied with burr-free square-cut or beveled ends, protective plastic end caps, and full ink-jet or laser stenciling showing heat number, grade, dimensions, and governing ASTM/ASME/EN specification.

PAGE 04 · SEAMLESS STAINLESS STEEL TUBES DIVISION (PART 2)

# Coil Tubing, Instrumentation Lines, Heat Exchanger Tubes & Master Capabilities

For continuous long-distance fluid routing, high-pressure compression-fitting instrumentation, and shell-and-tube thermal bundles, AV-Tech Tube Industry engineers specialized cold-drawn tubing with controlled hardness and bright annealed (BA) or pickled-passivated finishes.

## Coil Form, Instrumentation & Heat Exchanger Tubing Specifications

| Product Line | Dimensional Range & Codes | Engineering Design & Operational Benefits |
|---|---|---|
| Seamless Coil Form Tubes | **Length:** Up to 250 meters<br>**Core Grades:** SS 304L / 316L<br>**Supply:** Heavy-duty reels | Continuous seamless lengths wound into compact coils for instrumentation, hydraulic control, chemical injection, and oil & gas umbilical service — eliminating up to 90% of inline couplings and potential leak joints. |
| Instrumentation Tubing | **OD:** 1/8" – 2" (3.00 – 50.80 mm)<br>**WT:** 0.60 – 3.00 mm<br>**Standard:** ASTM A269 / A213 | Small-diameter cold-drawn tubing manufactured to tight OD, ovality, eccentricity, and Rockwell hardness (HRB ≤ 80/90) tolerances for leak-free twin-ferrule compression fittings on pressure, impulse, and analyzer lines. |
| Heat Exchanger Tubing | **OD:** 6.35 – 50.80 mm<br>**Standard:** ASTM A213 / SA213<br>**Finish:** BA or Pickled-Passivated | Cold-drawn seamless tubing engineered for shell-and-tube heat exchangers, condensers, evaporators, and reboilers, featuring smooth internal surfaces for low fouling and high resistance to chloride pitting. |

## Master Manufacturing Range, Alloy Grades & International Standards (avtechtube.com/products)

| Specification Parameter | Certified Capability across AV-Tech Tube Production Lines |
|---|---|
| **Overall Outer Diameter (OD)** | 3.00 mm to 60.30 mm (Imperial: 1/8" to 2-3/8" OD; Pipe sizes: 1/4" NB to 4" NB) |
| **Wall Thickness (WT) Range** | 0.60 mm to 8.00 mm (Schedules SCH 5S, 10S, 40S, 80S, 160S & custom metric gauges) |
| **Delivery Lengths & Forms** | Straight random or cut lengths up to 12 meters; U-bends to custom radii; Continuous coils up to 250 meters |
| **Austenitic Stainless Grades** | TP 304, 304L, 304H, 316, 316L, 316Ti, 317L, 321, 321H, 347, 347H, 904L (UNS N08904) |
| **Duplex & Specialty Alloys** | Duplex 2205 (UNS S31803 / S32205), Super Duplex 2507 (UNS S32750), 6% Moly Alloys, 316L VAR / VIM-VAR |
| **Governing ASTM / ASME Codes** | ASTM A213 / ASME SA213, ASTM A269, ASTM A270, ASTM A312 / SA312, ASTM A268, ASTM A789 / A790 |
| **European & Specialty Codes** | EN 10216-5, DIN 17456 / 17458, ASME BPE, SEMI F19 / F20, EN 10204 3.1 & 3.2 Certification |
| **Surface Finishes Available** | Mill finish, acid pickled & passivated, bright annealed (BA), mechanically polished (180–600 grit), electropolished (EP) |

**Seamless Tube Division:** Coil Form Tubes · Instrumentation Tubing · Heat Exchanger Tubing

PAGE 05 · HIGH PURITY TUBING DIVISION

# Ultra-High Purity, Electropolished & Hygienic Tubing for Cleanroom Service



AV-Tech High Purity Division: Mirror-bright internal bores, precision U-bends, helical corrugated profiles, and orbital-weld-ready tubing.

## High Purity Division Product Lines (avtechtube.com/products)

| High Purity Product | Surface & Governing Code | Cleanroom Processing, Chemistry & Target Service |
|---|---|---|
| Ultra-High Purity (UHP) Tubing | **Ra 0.127 – 0.25 µm** ASTM A-380 / CGA G4.1 / ASTM G93 | Manufactured in 316L and 316L VAR / VIM-VAR with electropolished bore, UHP nitrogen dried, end-capped, and double-bagged inside a **Class 4 cleanroom**. |
| Electropolished (EP) Tubing | **Ra ≤ 0.25 µm** 10–15 µm Metal Removal | Electrochemical polishing levels micro-peaks to achieve a mirror-bright bore with an enriched chromium-to-iron passive layer that eliminates particle trapping. |
| Bioprocessing (BPE) Tubing | **ASME BPE 316L** SF1 (Ra ≤ 0.51 µm) & SF4 (Ra ≤ 0.38 µm) | Tightly controlled sulphur chemistry (**0.005–0.017%**) ensures consistent penetration during automated orbital welding in biopharma WFI and bioreactor lines. |
| Semiconductor (SEMI) Tubing | **SEMI F19 / F20** SEMI F60 / F72 / F73 Tested | 316L tubing engineered for ultra-pure specialty gas and chemical distribution in semiconductor wafer fabs, backed by surface composition characterization. |
| Sanitary / Hygienic Tubing | **ASTM A270 & 3-A** Crevice-Free Polished Bore | Hygienic tubing designed for Clean-in-Place (CIP) and Steam-in-Place (SIP) food, dairy, beverage, and personal care processing lines. |
| Passivated Tubing | **ASTM A380 / A967** Nitric or Citric Acid Treatment | Chemical passivation removes exogenous free iron and strengthens the protective chromium oxide film without altering tube dimensions. |

## Cleanroom Analytical Verification & Packaging Protocol

High-purity consignments are verified via 100% digital boroscope inspection, profilometry, particle counting, moisture and total hydrocarbon analysis, and XPS/AES & SEM surface characterization reported on EN 10204 3.1 certificates.

**High Purity Division:** UHP · EP · ASME BPE · SEMI · Sanitary · Passivated

PAGE 06 · GLOBAL INDUSTRY APPLICATIONS

# Ten Critical Sectors, Ten Different Reasons the Tube Must Not Fail

AV-Tech Tube Industry partners with EPC contractors, equipment manufacturers, and end-user facilities across ten high-consequence industries where weld-zone failure, contamination, or corrosion downtime is unacceptable.

## Ten-Sector Engineering & Product Application Matrix (avtechtube.com/industries)

| No. | Industry Sector | Typical Products & Grades | Operating Environment & Service Demands |
|---|---|---|---|
| 01 | **Oil, Gas & Petrochemical** | Straight tubes, Exchanger & Coils (316L, Duplex 2205/2507) | High-pressure process piping, offshore control lines, and refinery bundles exposed to sour gas and chloride attack. |
| 02 | **Semiconductor Manufacturing** | UHP, EP & SEMI F19/F20 tubing (316L VAR / VIM-VAR) | Ultra-pure specialty gas and chemical distribution in wafer fabs requiring Ra 0.127–0.25 µm bores and zero outgassing. |
| 03 | **Food, Beverage & Dairy** | ASTM A270 / 3-A Sanitary & Corrugated tubes (304L/316L) | Crevice-free hygienic interiors for CIP/SIP sanitation and corrugated tubular heat exchangers for high-viscosity media. |
| 04 | **Pharmaceutical & Biotech** | ASME BPE SF1/SF4 & Passivated tubing (316L controlled S) | Water-for-Injection (WFI) loops, pure steam, and bioreactors demanding repeatable orbital welds and zero microbial harborage. |
| 05 | **HVAC & Refrigeration** | Corrugated, Coil & Heat Exchanger tubes (304L/316L) | Chillers, heat pumps, condensers, and evaporators leveraging 2–3x higher convective thermal transfer efficiency. |
| 06 | **Aerospace & Defence** | Precision thin-wall Instrumentation tubing (304L/316L/321) | High-pressure hydraulic actuation, fuel delivery, and environmental control lines under severe vibration and thermal cycling. |
| 07 | **Automotive & Transport** | Cold-drawn Seamless & Corrugated tubing (304L/316L/321) | High-pressure fuel injection rails, EGR coolers, Hydrogen/CNG lines, and corrosion-resistant brake and fluid routing. |
| 08 | **Chemical Processing** | Exchanger, U-Tubes & Straight pipes (316L, 904L, Duplex) | Reactors, reboilers, and acid condensers handling aggressive chlorides, sulfuric/phosphoric acids, and caustic media. |
| 09 | **Power Generation** | ASTM A213 U-Tubes, Boiler & Instrumentation tubes | High-pressure feedwater heaters, surface condensers, auxiliary steam boilers, and turbine sensing instrumentation. |
| 10 | **Construction & Architecture** | Mechanically Polished Straight & Structural seamless tubes | Coastal architectural facades, structural balustrades, district heating/cooling networks, and water treatment plants. |

**Industries Overview:** www.avtechtube.com/industries · Custom Alloy & Dimensional Engineering Available for All 10 Sectors

PAGE 07 · QUALITY, COMPLIANCE & SUSTAINABILITY

# Certified, Traceable & Documented Quality Systems Across Every Consignment

At AV-Tech Tube Industry, quality assurance governs every stage from raw material hollow selection to final seaworthy or cleanroom dispatch. Our management systems are independently audited and certified to **ISO 9001:2015, ISO 14001:2015, and ISO 45001:2018**.

## Non-Destructive, Mechanical & Analytical Testing Protocols (avtechtube.com/quality)

| Inspection / Test Category | Standard Verification Procedure & Acceptance Criteria |
|---|---|
| **Chemical & Alloy Verification** | 100% Positive Material Identification (PMI) via optical emission / XRF spectrometer on incoming hollows and finished tubes; controlled sulphur verification for ASME BPE heats. |
| **Non-Destructive Testing (NDT)** | 100% Eddy-Current testing (ASTM E426) and/or Ultrasonic testing (ASTM E213) to detect longitudinal or transverse discontinuities across the full tube length. |
| **Pressure Integrity Testing** | 100% Hydrostatic pressure testing or pneumatic underwater air-test in accordance with ASTM A213, A269, A312, and EN 10216-5 standards. |
| **Mechanical & Metallurgical Tests** | Tensile strength, yield strength, elongation, Rockwell/Vickers hardness, flattening test, flaring test, reverse bend test, and Intergranular Corrosion (IGC) per ASTM A262 Practice E. |
| **Surface & Dimensional Audit** | Laser micrometer OD/WT gauging, digital boroscope internal weld/bore inspection, and calibrated stylus profilometry (Ra µm) for EP, BPE, and UHP tubing. |
| **Certification & Traceability** | Full heat-number marking along each tube and **EN 10204 3.1 / 3.2 Mill Test Certificates (MTC)** welcoming third-party inspection (TPI) agencies (Lloyd's, TUV, Bureau Veritas, DNV, SGS). |

## Independently Assessed ISO Management System Certifications

| ISO Standard | Certified Management System Scope | Certificate No. | Validity Window |
|---|---|---|---|
| **ISO 9001:2015** | Quality Management System (Cold-Finish Seamless Tubes & Pipes) | **2026040218** | 02 Apr 2026 – 01 Apr 2029 |
| **ISO 14001:2015** | Environmental Management System | **2026040220** | 02 Apr 2026 – 01 Apr 2029 |
| **ISO 45001:2018** | Occupational Health & Safety Management System | **2026040219** | 02 Apr 2026 – 01 Apr 2029 |

## Sustainability & Product Lifecycle Responsibility (avtechtube.com/sustainability)

Stainless steel seamless tubing is 100% recyclable and engineered for multi-decade service life. Through our ISO 14001 and ISO 45001 systems, AV-Tech Tube Industry enforces closed-loop pickling acid neutralization, water conservation, energy-efficient bright annealing, and zero-harm shop-floor ergonomics.

**Quality & Sustainability:** www.avtechtube.com/quality · www.avtechtube.com/sustainability

PAGE 08 · GLOBAL CONTACT DIRECTORY, ORDERING PARAMETERS & EXPORT LOGISTICS

# Connect with AV-Tech Tube Industry for Technical Enquiries & Global Supply

## Corporate & Manufacturing Works Contact Directory (avtechtube.com/contact)

| Department / Office | Direct Contact Details | Role & Enquiry Scope |
|---|---|---|
| **Manufacturing Works & Head Office** | Plot No 13, Survey No. 1540, Near Hansalpur GIDC, Viramgam - Surendranagar Highway, Viramgam, Ahmedabad, Gujarat - 382150, India | Cold-drawing mill, bright annealing, U-bending, electropolishing, quality lab & global dispatch hub. |
| **Director Office** | **Email:** mineshvyas@avtechtube.com<br>**Phone:** +91 98982 32640 | Strategic partnerships, OEM/EPC frame agreements, and custom tube development. |
| **Marketing & Exports** | **Email:** marketing@avtechtube.com<br>**Phone:** +91 76002 65763 | International RFQs, distributor enquiries, pricing, lead times, and technical submittals. |
| **Factory Operations** | **Email:** info@avtechtube.com<br>**Phone:** +91 98982 32679 | Production tracking, third-party inspection (TPI) scheduling, and logistics coordination. |

## How to Specify — Technical Quotation & Ordering Parameters

Share your specification across the six parameters below — our engineering and commercial team will revert promptly with manufacturing feasibility, production lead time, and documentation scope:

| Specification Parameter | Details to Include in Your Technical Enquiry (Send Enquiry) |
|---|---|
| **01 · Material Grade & Chemistry** | Austenitic (304/304L, 316/316L, 321, 347, 904L), Duplex (2205/2507), 316L VAR/VIM-VAR, or controlled sulphur (0.005–0.017%). |
| **02 · Dimensions & Schedule** | Outer Diameter (3.00–60.30 mm or 1/8"–2" OD), Wall Thickness (0.60–8.00 mm), or Nominal Pipe Size (1/4"–4" NB, SCH 5S–160S). |
| **03 · Tube Form & Length** | Straight lengths (up to 12 m), helical Corrugated profile, cold-bent U-Tubes (with bend radius & leg length), or Coils (up to 250 m). |
| **04 · Governing Standard** | ASTM A213/SA213, A269, A270, A312/SA312, A268, A789/A790, EN 10216-5, DIN 17456/17458, ASME BPE, or SEMI F19/F20. |
| **05 · Surface Finish & Cleaning** | Pickled & passivated, bright annealed (BA), mechanically polished, electropolished (Ra ≤ 0.25 µm), or Class 4 UHP cleanroom bagged. |
| **06 · Certification & Packaging** | EN 10204 3.1 / 3.2 MTC, third-party inspection (TPI), seaworthy wooden boxes, steel-strapped bundles, or cleanroom $N_2$-purged sleeves. |

AV-TECH TUBE INDUSTRY

LET'S BUILD THE FUTURE TOGETHER

Precision cold-finish seamless stainless steel tubes & pipes manufactured in Viramgam, Gujarat, India · Web: www.avtechtube.com · ISO 9001 / 14001 / 45001

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